FEYMAN

FMI IMU AHRS

Software Integration Guide

AHRS output, serial protocols and calibration

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Software Reference: Feyman AHRS Tool 0.1.8

Interface: UART, AT command set

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Revision History

Version	Date	Change summary
V1.0	2026-10-15	Source device-protocol reference for Feyman AHRS Tool 0.1.8.
V1.1	2026-10-16	Reformatted to the FMI application-note style; added document navigation and reorganized integration and calibration guidance.

CHANGES IN V1.1 Added a dedicated cover, revision history, clickable contents and PDF bookmarks. Standardized section hierarchy, protocol tables, code examples, notes and page footers. Reorganized connection and calibration steps for integration use. Protocol values and firmware-specific qualifications are retained from V1.0.

APPLICABILITY This document describes AHRS serial output, message protocols and calibration behavior used by Feyman AHRS Tool 0.1.8 (2026-10-15). V1.1 is the document version; confirm the firmware actually loaded in the module with AT+VERSION.

Firmware and model scope

Commands, payloads and calibration behavior may vary by model and firmware. Read the module version with `AT+VERSION` and validate the actual hardware before production integration. Applicable model numbers are not specified in the supplied AHRS reference.

The source reference reports differences from an earlier vendor guide in working-mode values, calibration commands, frame lengths, thresholds and status bits. Use the AHRS definitions in this guide for the referenced firmware. The INS guide is used as a layout reference only.

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READING THE TABLES All binary byte offsets in this guide start at zero. $GPCAL field indices also start at zero, including the $GPCAL header. Message IDs are case-sensitive.

1 Introduction and Connection

This guide describes serial measurement output, magnetometer calibration, system calibration and the AT commands used by Feyman AHRS Tool 0.1.8. It is intended for developers integrating the referenced AHRS firmware into a host application.

1.1 Serial settings and stream handling

Setting	Value
Baud rate	115200 bit/s
Data format	8 data bits, no parity, 1 stop bit
Flow control	None
AT command encoding	Printable ASCII; terminate each command with CRLF (bytes 0D 0A).
Receive stream	Text lines and binary frames may be interleaved. A serial read is not necessarily a complete message.

Table 1. Serial settings used by the AHRS tool.

1.2 Connection sequence

Use the following sequence to identify the device UART, select AHRS mode and enable measurement output. Replace N with the device UART reported by the module.

```
1  AT+THIS_PORT
   Wait for both OK and This Port is UartN.
2  AT+WORK_MODE=2
   Continue after OK.
3  AT+AHRS_OUTPUT=UARTN,ON
   Enable output on the identified device UART.
```

UART NUMBER N identifies a UART on the device. It is not the computer COM-port number. To stop output without closing the computer port, send AT+AHRS_OUTPUT=UARTN,OFF.

Read a nonempty `Version:` response to `AT+VERSION`. Use `AT+READ_PARA=ALL` to retain the original parameter text for diagnosis and traceability.

PARAMETER PERSISTENCE Before issuing AT+SYSTEM_RESET, allow about 3 seconds for changed parameters to auto-save.

2 Configuration Commands

Send the command strings below as printable ASCII followed by CRLF. The overview records the completion conditions explicitly described in the source reference; do not assume that an arbitrary OK completes every operation.

Command	Purpose / completion condition
`AT+THIS_PORT`	Query the device UART; wait for both OK and This Port is UartN.
`AT+WORK_MODE=2`	Select AHRS mode during connection initialization; continue after OK.
`AT+AHRS_OUTPUT=UARTN,ON`	Enable AHRS output on device UART N.
`AT+AHRS_OUTPUT=UARTN,OFF`	Disable AHRS output on device UART N; keep the host port open.
`AT+VERSION`	Read a nonempty Version: reply.
`AT+READ_PARA=ALL`	Read parameters; retain the original device text.
`AT+MAG_CALI_CFG=2D / 3D`	Choose 2D (4-parameter) or 3D (8-parameter) magnetic calibration. Send one choice.
`AT+MAG_CALI_MANU=START`	Start a magnetometer calibration run.
`AT+MAG_CALI_MANU=STOP`	Stop the run without saving its result.
`AT+MAG_CALI_MANU=SAVE`	Attempt to save after the current-run indicator checks pass.
`AT+CALIBRATE_MODE=STEP1`	Begin the bottom-face and side-face static system calibration step.
`AT+CALIBRATE_MODE=STEP2`	Begin the subsequent six-face static step.
`AT+SYSTEM_RESET`	Reboot; first allow about 3 seconds for parameter auto-save.

Table 2. AHRS command set used by Feyman AHRS Tool 0.1.8.

Command scheduling and confirmation

Manual commands queue for transmission without waiting for a reply and without interrupting an active automatic command. Automatic commands check their corresponding replies. A system calibration step is confirmed by a matching `$GPCAL` line, not by an arbitrary OK.

FIRMWARE UPDATE Firmware transfer uses a separate APP/BOOT upgrade flow. Do not send upgrade commands from the manual command input. The upgrade protocol is outside this guide.

3 Common Binary Frame

The current `fmiA` and `fmib` messages use little-endian multibyte values. The payload-length field counts payload bytes only. The four-byte message ID is case-sensitive.

3.1 Frame layout and CRC

Region	Size	Content
Header	4 bytes	ASCII message ID: fmiA or fmib.
Payload length	2 bytes	Little-endian uint16; payload size only.
Payload	As specified	Device time and measurement / calibration values.
CRC	2 bytes	CRC16/XMODEM of the payload only; least-significant byte first.

Table 3. Common frame envelope for current AHRS messages.

```
frame_bytes = payload_bytes + 8
crc_initial = 0x0000
crc_polynomial = 0x1021
crc_test("123456789") = 0x31C3
```

CRC INPUT Exclude the header, length field and CRC field from the CRC input. Validate the payload bytes only. Current AHRS framing does not use the legacy additive checksum or the legacy ed tail.

3.2 Receiver handling

Buffer incomplete data across reads and handle multiple messages received in one read. Route text lines and binary frames separately. After an invalid length or CRC, resynchronize the stream before accepting another frame.

Message	Payload bytes	Complete frame bytes
`fmiA`	32	40
`fmib, current`	44	52
`fmib, old logs`	40	48

Table 4. Message lengths. Each fmib variant must pass its own length and CRC checks.

Unknown messages with valid length and CRC are skipped without producing a plot or export row. Keep raw logs when investigating an unsupported message or a firmware difference.

4 AHRS Message fmiA

Header: ASCII `fmiA`. Payload: 32 bytes. Complete frame: 40 bytes. All offsets below are zero-based; all multibyte values are little-endian.

4.1 Fields and physical-unit conversion

Offset	Bytes	Field	Type
`0`	4	Header: fmiA	`ASCII`
`4`	2	Payload length: 32	`uint16`
`6`	4	Device time, milliseconds	`uint32`
`10, 12, 14`	2 each	Acceleration X, Y, Z	`int16`
`16, 18, 20`	2 each	Angular velocity X, Y, Z	`int16`
`22, 24, 26`	2 each	Magnetic field X, Y, Z	`int16`
`28, 30, 32`	2 each	Roll, Pitch, Yaw	`int16`
`34`	4	status	`uint32`
`38`	2	Payload CRC16	`uint16`

Table 5. fmiA frame layout. Acceleration X is signed, like Y and Z.

```
time_s = time_ms / 1000.0
acc_g = raw_acc * 8.0 / 32000.0
gyro_deg_s = raw_gyro * 500.0 / 32000.0
mag_uT = raw_mag * 500.0 / 32000.0
angle_deg = raw_angle * 180.0 / 32000.0
```

time_ms is device time in milliseconds; raw_* are signed sensor values. Outputs use seconds, g, deg/s, microtesla and degrees.

4.2 Status bits

Mask	Meaning for the referenced firmware
`0x00000001`	Magnetic field available; magnetic-field indicator in 3D attitude.
`0x00000002`	Magnetic heading valid; magnetic-heading indicator in 3D attitude.

Table 6. Confirmed low-bit definitions. Test each bit independently.

```
mag_available = (status & 0x01) != 0
mag_heading_valid = (status & 0x02) != 0
```

Bit 0x01 follows `mag_manager.IsAvailable()`. Bit 0x02 is set when the last magnetic update was less than 60 seconds ago; it does not alone prove calibration completion. Values 0x00, 0x01, 0x02 and 0x03 mean neither, field only, heading only and both, respectively. Display received bits even when 0x02 occurs without 0x01. Other bits are unconfirmed: retain the full raw status for export and debugging. These meanings do not apply to legacy status fields.

5 Magnetometer Calibration

5.1 Calibration frame fmib

The current frame uses ASCII `fmib`, a 44-byte payload and a 52-byte complete frame. Older logs may use a 40-byte payload and a 48-byte complete frame. Each variant requires its own valid length and payload CRC.

Offset	Bytes	Field / type
`0`	4	Header: fmib, ASCII
`4`	2	Payload length: 44; 40 in old logs, uint16
`6`	4	Device time in milliseconds, uint32
`10 to 38`	4 each	Indicators 1 to 8, float32; offsets advance by 4.
`42`	4	Residual, float32
`46`	4	Reserved uint32 in current frame; absent in old frame.
`50 / 46`	2	CRC16: offset 50 current; offset 46 in old frame.

Table 7. fmib layout. All multibyte fields use little-endian byte order.

5.2 Thresholds and save conditions

Indicator	1	2	3	4	5	6	7	8
Threshold	1	1	1	0.05	0.05	0.05	0.02	0.02
2D check	Yes	Yes	No	Yes	No	No	Yes	No
3D check	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Table 8. Every active indicator must be strictly below its threshold; equality does not pass.

1. Choose the calibration configuration with `AT+MAG_CALI_CFG=2D` or `AT+MAG_CALI_CFG=3D`. 2D uses 4 parameters; 3D uses 8 parameters.

2. Send `AT+MAG_CALI_MANU=START` and evaluate indicator values received during this run. Earlier values cannot authorize saving.

3. When all active checks pass, send `AT+MAG_CALI_MANU=SAVE` to attempt to save. To end the run without saving, use `AT+MAG_CALI_MANU=STOP`.

DATA SOURCES The magnetic sample trace uses fmiA magnetic X/Y/Z values. fmib carries the calibration indicators and residual. Do not interpret the fmib indicators as magnetic samples.

6 System Calibration Message $GPCAL

`$GPCAL` is a comma-separated text line, not a binary `fmiA` or `fmib` frame. Field indices start at zero, including the header. The final field may carry a * suffix. The current tool reads fields 2 to 5 and 7 to 13; retain the positions of the other fields.

Field	Content	Meaning / values
`0`	Header	$GPCAL
`1`	Raw device time	Original device time field.
`2`	Average temperature	Current average temperature; -1 may indicate unfinished warm-up.
`3`	Static-stage temperature	Temperature for the static stage.
`4`	Acceleration standard deviation	Static acceleration standard deviation.
`5`	Gyroscope standard deviation	Static gyroscope standard deviation.
`6`	Reserved	Retain this field position.
`7`	Calibration accuracy	Standard deviation; may be -1 until complete.
`8`	Calibration step	STEP1 or STEP2; the tool also accepts 1 or 2.
`9`	Phase	0 warm-up; 1 static capture; 2 STEP1 complete; 3 STEP2 capture; 4 all complete.
`10`	Sample count	Number of samples reported by the device.
`11`	Operator instruction	Step-specific meaning; see Section 7.
`12`	Failure reason	0 means no reported reason; other codes are listed in Section 7.2.
`13`	Result	0 failed; 1 running; 2 succeeded.

Table 9. $GPCAL fields. Temperature and deviation units are not specified in the supplied AHRS reference.

COMPLETION Use a matching $GPCAL report to confirm the active calibration step. The tool requires STEP1 phase 2 before STEP2 is allowed. Result 2 is required to confirm overall success.

Preserve the complete original line when logging calibration progress or diagnosing a failure. Do not apply the binary payload-length or CRC16 envelope to this text message.

7 System Calibration Procedure

7.1 STEP1 and STEP2

STEP1 calibrates the IMU-to-case mounting angle using the bottom face and a side face. STEP2 then captures six static faces. Follow the device prompts and keep the device still whenever instructed.

```
1  AT+CALIBRATE_MODE=STEP1
   Follow the bottom-face and side-face prompts.
   Require a matching STEP1 report with phase 2.
2  AT+CALIBRATE_MODE=STEP2
   Complete the six static faces as prompted.
   Require result 2 to confirm overall success.
```

Step	Field 11	Operator action / state
STEP1	`0`	Hold the current face still.
STEP1	`1`	Bottom face is complete; turn to a side face.
STEP1	`2`	Bottom and side are both complete.
STEP2	`0`	Hold still.
STEP2	`1`	Turn to the next face.

Table 10. Device-specific operator instructions in $GPCAL field 11.

7.2 Failure reasons

Field 12	Reported reason
`0`	No reported reason
`1 / 2`	Result verification / parameter storage
`3 / 4 / 5`	Excessive temperature rise / calculation error / abnormal data
`6 / 8`	Residual checks
`7 / 9`	Variance checks
`10 / 11`	Insufficient samples / timeout

Table 11. Failure codes reported in $GPCAL field 12.

DIAGNOSIS Interpret the failure code together with the step, phase, result and original $GPCAL line. A zero failure reason alone is not proof of completion.

8 Legacy Logs and Integration Checks

8.1 Legacy replay scope

The AHRS tool can replay the older records below. Their additive checksum and time representation differ from the current AHRS frames. Physical units that have not been confirmed are displayed as raw values.

Message	Complete record bytes	Handling
`fmim`	52	Legacy additive checksum and legacy time representation.
`fmig`	95	Legacy additive checksum and legacy time representation.
`fmin`	100	Legacy additive checksum and legacy time representation.

Table 12. Legacy records supported for replay by the referenced tool.

PROTOCOL SEPARATION Do not reuse legacy time, units or status semantics for fmiA. Do not apply the current fmiA status-bit definitions to legacy status words. Firmware transfer uses a separate APP/BOOT protocol and remains outside this measurement-frame guide.

8.2 Integration checks

Use these checks when implementing the behavior described in this guide. Confirm the actual device responses and retain raw logs for any new model or firmware.

Area	Check
Connection	Confirm 115200 bit/s and 8N1 with no flow control. Match device UART N to the THIS_PORT reply.
Stream parser	Exercise split reads, consecutive frames, interleaved text, invalid lengths and invalid CRCs.
Binary decode	Verify little-endian values, signed int16 measurements, payload-only CRC and zero-based offsets.
Status	Test the low two bits independently, including 0x02 without 0x01; retain the full raw status.
Magnetic calibration	Check the 2D / 3D indicator selection, strict thresholds and exclusion of stale values.
System calibration	Require matching $GPCAL step reports, STEP1 phase 2 and result 2 for overall success.
Traceability	Retain raw logs, firmware-version text and original parameters. Validate each new model or firmware on hardware.

Table 13. Integration review checklist derived from the supplied AHRS reference.